

Vulnerability Assessment Report

27th June 2025

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

The database server is valuable to the business because they store information here, and since many of the employees are remote, they query and request data from the server regularly to find potential customers. The server connects to other servers on the network which also makes it valuable. It is important for the business to secure the data on the server to minimize potential risks and ensure confidentiality, integrity, and availability of information. If the server was disabled it would severely impact the business since they rely on it regularly to find potential customers. It would disrupt their day to day business activities.

- *How is the database server valuable to the business?*
- *Why is it important for the business to secure the data on the server?*
- *How might the server impact the business if it were disabled?*

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtain sensitive information via exfiltration	3	3	9

<i>Customer</i>	<i>Alter/Delete critical information</i>	<i>1</i>	<i>3</i>	<i>3</i>
<i>Hacker</i>	<i>Conduct denial of service attacks</i>	<i>2</i>	<i>3</i>	<i>6</i>

Approach

I selected 2 human threat sources and 1 technical threat source because I was able to see these sources based on the information I've gathered, the purpose, and the system description. The threat events were selected to align with the information in the purpose. I think these are all significant business risks because if threat actors found out these vulnerabilities, they would most definitely be exploited by threat actors. Most of the employees are remote and regularly query and request data so a denial of service or alter/delete critical information would greatly impact the business. Since it's a public database server that shouldn't be public, it's always susceptible to theft of sensitive information.

Remediation Strategy

Use public key infrastructure(PKI) to address exfiltration of sensitive information. Implement AAA framework to address the alter/delete critical information by monitoring access logs to see who and when the system was accessed. Defense in depth can be used to mitigate a denial of service attack.